

System Design Document

SentinelLite: A Lightweight SIEM with Rule-Based Detection and ML Anomaly Scoring

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Course	Information Security (D1)
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Architecture

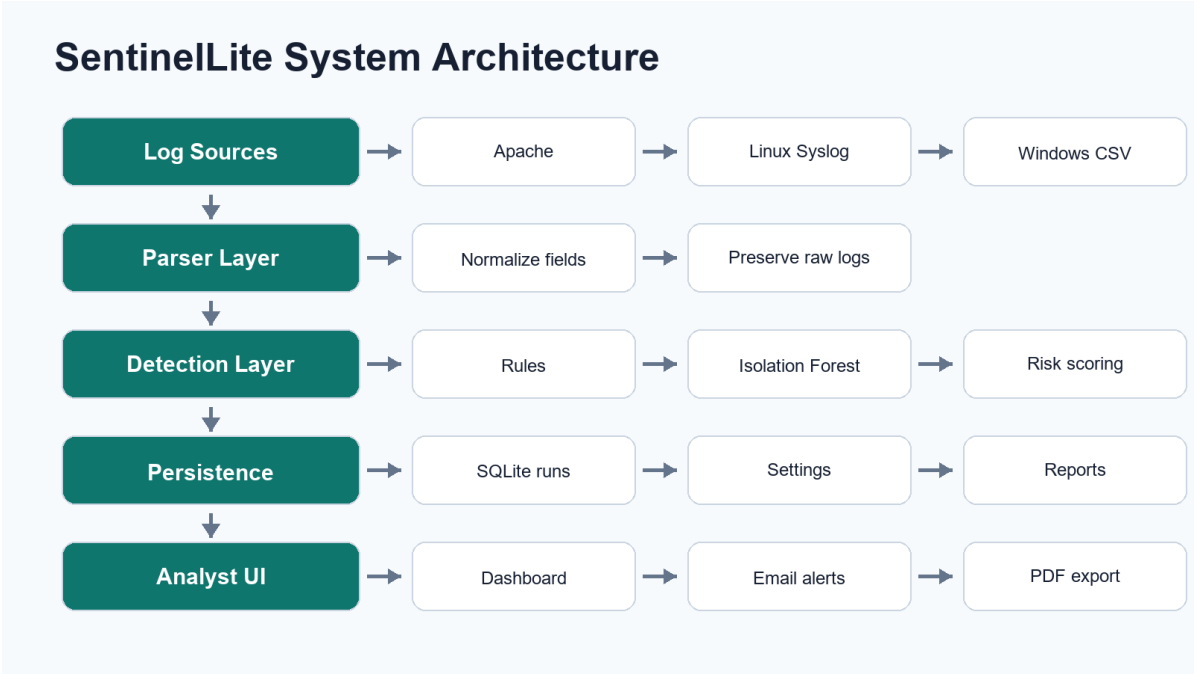


Fig. 1. SentinelLite system architecture.

The architecture separates ingestion, normalization, detection, anomaly scoring, persistence, and analyst interaction. This separation improves testability and makes each component easy to defend in viva.

Data Flow

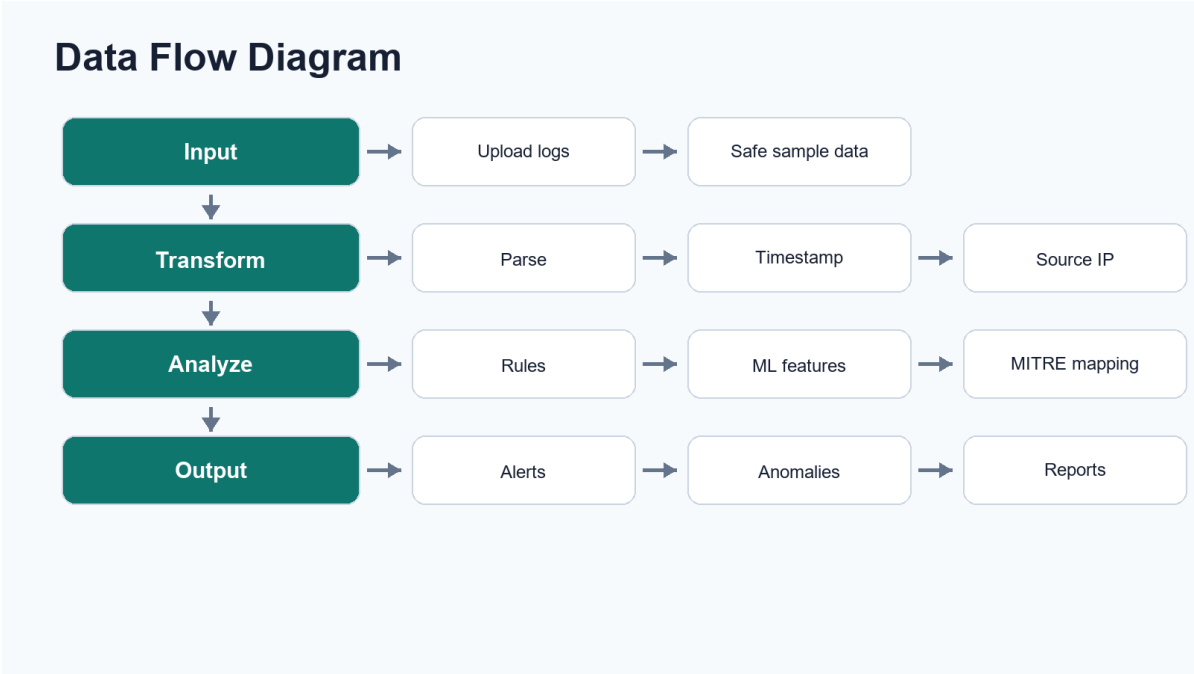


Fig. 2. SentinelLite data-flow diagram.

Threat Model

Threat	Surface	Control
Malicious log payload text	Uploaded log files	Treat logs as text; do not execute payload content.
Credential attack	Authentication logs	Failed-login burst and success-after-failure rules.
Reconnaissance	Web/firewall logs	404 enumeration, scanner UA, and port-scan rules.
Evidence tampering	Windows audit logs	Event ID 1102 detection and PDF reporting.

Technology Justification

Python and Pandas provide quick, readable parsing and aggregation. SQLite satisfies the brief's local storage option with low setup cost. Scikit-learn Isolation Forest provides unsupervised anomaly scoring without paid APIs. Streamlit gives an interactive dashboard with simple deployment. ReportLab creates PDF evidence reports.